=== UNIT 4: ANALYSIS OF PLANE TRUSSES & STRUCTURES ===

>> 4.1 Engineering Trusses and Determinacy Criteria

A truss is an engineering structure composed of slender structural members joined together at their end terminals to form a rigid stationary framework capable of spanning large distances (roof trusses, railway bridges, crane booms).
* Classification by Internal Determinacy: Let m = total number of members, j = total number of pin joints:
- Perfect (Statically Determinate) Truss: m = 2*j - 3. The truss is rigid and all member forces can be uniquely solved using static equilibrium equations alone.
- Deficient (Hypostatic / Unstable) Truss: m < 2*j - 3. Lacks sufficient members to prevent collapse under general loading.
- Redundant (Statically Indeterminate) Truss: m > 2*j - 3. Possesses redundant members; cannot be solved by statics alone without deformation compatibility.

>> 4.2 Fundamental Assumptions in Plane Truss Analysis

To facilitate analytical solution, ideal truss analysis makes four classical idealizations:
* 1. All members are connected at their ends by frictionless smooth spherical pins or cylindrical hinges.
* 2. External loads and support reactions are applied strictly at the joints (nodes) only.
* 3. The centroidal longitudinal axes of all members meeting at a joint are concurrent at the joint pin center.
* 4. Self-weights of members are either negligible or split equally as concentrated half-weights applied at end joints.
* Consequence: Every member is a pure two-force member carrying either axial Tension (T, pulling away from joint) or axial Compression (C, pushing toward joint). Bending moments and shear forces are zero.

>> 4.3 Zero-Force Member Identification by Inspection

Structural efficiency is greatly expedited by identifying zero-force members prior to numerical computation:
* Rule 1 (Two-Member Joint): If only two non-collinear members meet at an unloaded joint without support reactions, the internal force in BOTH members must be ZERO: F1 = 0 and F2 = 0.
* Rule 2 (Three-Member Joint): If three members meet at an unloaded joint where two members are collinear, the third non-collinear member MUST carry ZERO force: F3 = 0, and F1 = F2.

* Rule 3 (Loaded Joint): If two non-collinear members meet at a joint subjected to an external load collinear with one member, the other non-collinear member carries ZERO force.

>> 4.4 Method of Joints

The Method of Joints isolates individual pin joints as free body diagrams of concurrent force systems:
* Step 1: Determine global external support reactions by treating the entire truss as a rigid body: Sigma Fx = 0, Sigma Fy = 0, Sigma M = 0.
* Step 2: Select a joint having at least one known force and NOT MORE THAN TWO unknown member forces.
* Step 3: Draw joint FBD, assuming unknown member forces are Tensile (arrows pointing away from the pin).
* Step 4: Apply concurrent equilibrium: Sigma Fx = 0, Sigma Fy = 0. A positive calculated value confirms tension (+T); a negative value denotes compression (-C).
* Step 5: Advance sequentially to adjacent joints until all member forces are computed.

>> 4.5 Method of Sections

When internal forces in only a few specific internal members are desired (e.g. central chord or diagonal web members of a bridge), the Method of Sections is far faster than the Method of Joints:
* Step 1: Calculate global external support reactions.
* Step 2: Pass an imaginary cutting section through the truss, cutting the members whose forces are required (cutting through NOT MORE THAN THREE members whose forces are non-concurrent).
* Step 3: Isolate one side of the severed truss (left or right) as a non-concurrent rigid body FBD.
* Step 4: Expose internal axial forces in cut members as external forces acting along their member lines of action.
* Step 5: Apply non-concurrent planar equilibrium: Sigma M_joint = 0 (taking moments about the intersection of two unknown forces eliminates both simultaneously), Sigma Fx = 0, Sigma Fy = 0.

>> 4.6 Comparison of Trusses, Frames and Machines

* Truss: Structure comprised entirely of two-force members carrying only axial tension or compression.
* Frame: Stationary structure containing at least one multi-force member (member subjected to 3 or more forces or transverse loads, carrying bending moments and shear forces) designed to support external loads.

* Machine: Structure designed to transmit and modify forces containing moving parts and multi-force members.

[EXAMPLE 1: Method of Joints on Triangular Cantilever Truss]

Problem Context: A cantilever truss consists of three joints: Wall supports A (fixed pin) and B (roller directly below A, vertical distance AB = 3.0 m). Joint C is located 4.0 m to the right of joint B. A vertical downward load of P = 24 kN is applied at joint C. Members are AC, BC, and vertical support line AB. Determine the forces in members AC and BC.

Solution Step 1: Geometry of members at joint C:

In right triangle ABC: Horizontal base BC = 4.0 m, Vertical side AB = 3.0 m.

Inclined tie member AC length = sqrt(4^2 + 3^2) = 5.0 m.

Inclination angle theta of AC with horizontal BC: cos(theta) = 4/5 = 0.80, sin(theta) = 3/5 = 0.60.

Solution Step 2: Analyze Joint C using equilibrium:

Only two unknown member forces meet at joint C: F_AC (pulling up and left at angle theta) and F_BC (acting horizontally along BC).

Assume F_AC is tensile (pointing away from C), F_BC is compressive (pointing toward C, or assume tensile to the left):

Vertical Equilibrium (Sigma Fy = 0 at C):

+ F_AC * sin(theta) - P = 0 => F_AC * 0.60 = 24 kN => F_AC = 24 / 0.60 = + 40.0 kN (Tension).

Horizontal Equilibrium (Sigma Fx = 0 at C):

- F_AC * cos(theta) - F_BC = 0 => F_BC = - F_AC * cos(theta) = - 40.0 * 0.80 = - 32.0 kN (Compression).

Solution Step 3: Verify Wall Support Reactions at A and B:

At A (pin): Member AC pulls joint A down and right with 40 kN => H_A = -32 kN (left), V_A = +24 kN (up).

At B (roller): Member BC pushes joint B left with 32 kN => Normal reaction H_B = +32 kN (right).

Conclusion: Member AC = 40.0 kN (Tension), Member BC = 32.0 kN (Compression).

[EXAMPLE 2: Method of Sections for a Pratt Bridge Truss]

Problem Context: A simply supported Pratt bridge truss has a span of 16 m with four equal panels of b = 4 m each and height h = 3 m. Vertical loads of 30 kN each act downward at lower panel joints L1, L2, L3. Determine the forces in top chord U1U2, diagonal U1L2, and bottom chord L1L2 by the Method of Sections.

Solution Step 1: Global Support Reactions at L0 and L4:

Total load = 30 + 30 + 30 = 90 kN. By symmetry: R_L0 = R_L4 = 90 / 2 = 45 kN (upward).

Solution Step 2: Pass cutting plane section 1-1:

Cut vertically through panel 2, severing top chord U1U2, diagonal U1L2, and bottom chord L1L2. Isolate the left part (joints L0, U1, L1).
Geometry of diagonal U1L2: horizontal run = 4 m, vertical rise = 3 m, length = 5 m.
sin(theta) = 3/5 = 0.60, cos(theta) = 4/5 = 0.80.
Solution Step 3: Determine Top Chord Force F_U1U2 using Moment Equilibrium about joint L2:
Passing moments about L2 eliminates both F_L1L2 and F_U1L2 (their lines pass through L2):
Sigma M_L2 = 0:
(R_L0 * 8.0 m) - (Load at L1 * 4.0 m) + (F_U1U2 * 3.0 m) = 0.
(45 * 8.0) - (30 * 4.0) + 3.0 * F_U1U2 = 0.
360 - 120 + 3.0 * F_U1U2 = 0 => 3.0 * F_U1U2 = - 240 => F_U1U2 = - 80.0 kN (Compression).
Solution Step 4: Determine Diagonal Force F_U1L2 using Vertical Equilibrium (Sigma Fy = 0):
Sigma Fy = 0:
R_L0 - (Load at L1) - F_U1L2 * sin(theta) = 0.
45 - 30 - F_U1L2 * 0.60 = 0 => 0.60 * F_U1L2 = 15 => F_U1L2 = + 25.0 kN (Tension).
Solution Step 5: Determine Bottom Chord Force F_L1L2 using Moment Equilibrium about joint U1:
Sigma M_U1 = 0 (joint U1 is at x = 4.0 m, y = 3.0 m):
(R_L0 * 4.0 m) - (F_L1L2 * 3.0 m) = 0.
(45 * 4.0) - 3.0 * F_L1L2 = 0 => 3.0 * F_L1L2 = 180 => F_L1L2 = + 60.0 kN (Tension).
Conclusion: F_U1U2 = 80.0 kN (Compression); F_U1L2 = 25.0 kN (Tension); F_L1L2 = 60.0 kN (Tension).

[EXAMPLE 3: Inspection and Identification of Zero-Force Members]

Problem Context: In a roof truss, joint D connects horizontal chord CD, vertical web member DF, and horizontal chord DE. Joint F has three members: vertical DF, and diagonals CF and EF forming a symmetric V-shape. No external load acts at joint D. Identify which member carries zero force and prove by joint equilibrium.
Solution Step 1: Examine Joint D:
Three members meet at joint D: CD and DE are collinear along the horizontal axis, and DF is perpendicular (non-collinear). No external load or support reaction acts at D.
Applying Sigma Fy = 0 at joint D: DF is the ONLY member possessing a vertical component.
Sigma Fy = F_DF = 0. Hence, F_DF = 0 (Zero-Force Member).
Applying Sigma Fx = 0 at joint D: - F_CD + F_DE = 0 => F_CD = F_DE (chord force is continuously transmitted).
Conclusion: Member DF is a zero-force member (F_DF = 0). It serves purely to provide lateral buckling restraint to the compression chord CD-DE.

[EXAMPLE 4: Analysis of a Howe Roof Truss Chord Forces]

Problem Context: A symmetric Howe roof truss of span L = 12 m and height H = 4 m carries a single ridge apex load P = 36 kN at joint C. Members are lower tie AB (span 12 m with central joint D), rafters AC and BC (length = sqrt(6^2 + 4^2) = 7.21 m), and central vertical tie CD = 4 m. Determine forces in all members.

Solution Step 1: Support reactions at A and B:

R_A = R_B = P / 2 = 36 / 2 = 18 kN (upward).

Solution Step 2: Joint D Analysis:

Members AD and DB are collinear horizontally. Member CD is vertical. Unloaded joint.

By Rule 2: Force in central vertical post F_CD = 0 (Zero-force member!).

F_AD = F_DB = F_tie.

Solution Step 3: Joint A Analysis:

cos(theta_A) = 6.0 / 7.21 = 0.8321, sin(theta_A) = 4.0 / 7.21 = 0.5547.

Vertical equilibrium: R_A + F_AC * sin(theta_A) = 0 => 18 + F_AC * 0.5547 = 0 => F_AC = - 32.45 kN (Compression).

Horizontal equilibrium: F_tie + F_AC * cos(theta_A) = 0 => F_tie = - (-32.45 * 0.8321) = + 27.0 kN (Tension).

Conclusion: Rafters AC and BC = 32.45 kN (Compression); Bottom tie AB = 27.0 kN (Tension); Vertical post CD = 0 kN.

>> 4.7 University Review & Examination Questions

* Q1: Differentiate comprehensively between the Method of Joints and the Method of Sections, detailing when each is preferred.

Answer: The Method of Joints isolates individual joints as concurrent force systems and solves unknown member forces sequentially using Sigma Fx = 0 and Sigma Fy = 0. It is preferred when internal forces in ALL members of a truss must be cataloged for complete structural fabrication. However, calculation errors at initial joints propagate to all subsequent nodes. The Method of Sections passes a cutting plane severing not more than three non-concurrent members, treating each isolated sub-structure as a rigid body and applying Sigma Fx = 0, Sigma Fy = 0, and Sigma M = 0. It is highly preferred when only specific internal chord or web members need verification, and errors do not propagate since each member can be independently solved using moment equilibrium.

* Q2: What are zero-force members, and why are they incorporated into engineering truss designs if they carry no force?

Answer: Zero-force members are structural elements in a truss that carry no axial tensile or compressive force under a specific static loading condition. Despite carrying zero force under normal primary loads, they are vital in real-world civil engineering structures because: (1) They provide essential lateral bracing to prevent long slender compression members from catastrophic elastic buckling. (2) They ensure stability and prevent structural collapse when loading conditions change (e.g. moving traffic loads, wind gusts, seismic shaking). (3) They maintain geometry during fabrication, lifting, transport, and erection.

>> 4.8 Key Takeaways and Revision Points

* Determinacy criterion for plane trusses: m = 2*j - 3.
* All truss members are idealized as two-force members carrying only axial tension or compression.
* Method of Joints uses concurrent equilibrium (Sigma Fx = 0, Sigma Fy = 0) at each node.
* Method of Sections uses non-concurrent equilibrium (Sigma M = 0) across an isolated section.